Riya Rampalli
rr3491@columbia.edu | GitHub

EDUCATION

MA. Columbia University, Ecology, Evolution, and Conservation Biology (Expected: May 2026)
Honor: Columbia Alumni Association (CAA) Scholar
Post-Bacc Certificate. Columbia University, Ecology, Evolution, and Conservation Biology (2022-2024)
BS. Mills College, Environmental Science (2018-2022)

RESEARCH EXPERIENCE

COLUMBIA UNIVERSITY - Department of Ecology, Evolution, and Environmental Biology
Graduate Student Researcher, The Eaton Lab, August 2024 - present
PI: Deren Eaton

- Investigating the genetic architecture associated with incompatibilities in hybrids, using dioecious Amaranth species, *Amaranthus tuberculatus* and *A. palmeri,* as model organisms.
- Generating PacBio HiFi sequencing long reads of pollen pools and leaf tissue for pure and hybrid species.
- Constructing genome-wide recombination maps and mapping transmission ratio distortion (TRD) using Python, CLI, and bash script.
- Creating *Distortopia,* an open-source computational tool hosted on GitHub, to be published for wider hybrid research use. *Distortopia* automates alignment and variant calling pipelines with improved reproducibility and scalability benefits.
- Preliminary analyses validated pipeline accuracy and scalability using a subset of test data

COLUMBIA UNIVERSITY - Mortimer B. Mind Brain Behavior Institute
Post-baccalaureate Student Researcher, The Bendesky Lab, October 2022 - May 2024
PI: Andrés Bendesky; Mentor: Wyatt Toure, PhD Candidate

- Used genetic, genomic, and molecular approaches to understand the mechanisms and genes underlying the physiological variation between two species of deer mice (*Peromyscus polionotus* and *P. maniculatus*) and their hybrid progeny.
- Assisted with mating monitoring, embryo collection, DNA/mRNA extractions, tissue collection, and the cross-sectioning of tissues.
- Early results show an increased dosage of X-linked genes were responsible for the smaller body sizes seen in hybrids.

LAWRENCE BERKELEY NATIONAL LABORATORY
Undergraduate Student Research Assistant, Mammalian Functional Genomics Laboratory, October 2021- May 2022
PI: Len Pennachio, Axel Visel, Jennifer Akiyama

- Assisted with comparative genomics and systems-based chromatin studies (ChIP-seq), genome engineering methods (CRISPR/Cas9), and transgenic reporter assays, which allowed identification and characterization of distant-acting transcriptional enhancers at a genomic scale.
- Annotated genotypes to determine differences in the genetic makeup of mice embryos and establish whether they were transgenic.
- Managed wet lab upkeep by autoclaving dishes, restocking buffer solutions, etc.

MILLS COLLEGE - Department of Biology
Undergraduate Student Researcher, Behavioral Ecology Lab, August 2021- May 2022
PI: Dr. Jennifer Smith

- Studied physiological stress responses in California Ground Squirrels across disturbed and undisturbed habitats by quantifying fecal glucocorticoid metabolites through extraction and ELISAs. Analyses showed ground squirrel response varies spatially to environmental stressors and influences their dietary intake.

SMITHSONIAN ENVIRONMENTAL RESEARCH CENTER (SERC)
NSF-REU Fellow, San Francisco State University, Summer 2020
PI: Dr. Gail Ashton

- Performed point counts of benthic marine invertebrates with GIMP and R on 15 x 15cm PVC settlement panels exposed to ~1m of water for 3 months in the coastal South Pacific. Performed ANOVAs and Tukey posthoc tests on Shannon diversity index. calculations to understand shifts in species composition and accumulation. Results suggested that predation intensity on marine invertebrates varies along a latitudinal gradient.

POSTERS

- Student Conference on Conservation Science - New York (SCCS-NY). **Rampalli R.,** Timerman, D., Eaton, D. "Genomic Insights into Hybridization in Dioecious *Amaranthus*."
- Society for the Study of Evolution: Evolution 2025. **Rampalli, R.**, Toure M.W., Bendesky, A. "Molecular Correlates of an X-linked Hybrid Incompatibility in *Peromyscus*"
- Genetic Society of America: The Allied Genetics Conference (TAGC'24). Toure M. W., **Rampalli R.**, Bendesky A. "Genetic and Molecular Bases of Hybrid Dysgenesis in Deer Mice." **Best Poster Award (one of several).**

AWARDS

- *Columbia Alumni Association Scholarship*, Columbia University (2025-2026) ~$5000
- *Arts & Sciences Graduate Council: Student Travel Grant*, Columbia University (Summer 2025) ~$500
- *Bill Dahl Graduate Student Research Award*, Botanical Society of America (Spring 2025) ~$1500
- *E3B Student Research Grant*, Columbia University (Spring 2025) ~$2000
- *Graduate Students of Arts & Sciences Matching Grant*, Columbia University (Spring 2025) ~$300
- *GSAS MA Scholarship Fund*, Columbia University (Fall 2024) ~$8000
- *Undergraduate Research Opportunities Program* (UROP) Grant, Mills College (2021) ~$250.00
- *Janet L. Holmgren Presidential Scholar*, Mills College (2018-2022) ~$40,000

WORK EXPERIENCE

NEW YORK ACADEMY OF SCIENCES
Afterschool STEM Mentoring Program (ASMP), Mentor Community Leader, Fall 2023 - Spring 2024
Coordinator: Danielle Mink-Bellizzi

- Facilitated onboarding, engagement, and communication for scientist-educator teams. Responsibilities included: (1) onboarding cohorts, (2) providing best practices for classroom engagement, (3) encouraging mentor engagement on the ASMP Slack channels, (4) supporting assigned scientist-educator teams, (5) checking in weekly with ASMP program teams, (6) identifying and supporting the development of "Mentor Spotlights", and (7) supporting in organizing teams when substitutes are needed.

FLINT BEACH OHANA, LLC
Summer Intern, Summer 2021

- Worked on forest land in non-Federal ownership (i.e., privately owned land) on the San Juan Islands. Assisted with trail management by clearing fallen debris and uprooting/removing invasive plant species. Cared for pasture-raised Romney Sheep and free-range chickens, including assisting in the vaccination of 30+ sheep for Clostridial C & D and Tetanus.

NEXTGEN POLICY
Policy Intern, Summer 2021
Manager: Ken Spence (Senior Policy Advisor)

- Advocated for criminal justice legislation by drafting legislative letters of support and opposition to California's speakers' offices (Senate/Assembly) and committees. Attended Board of State and Community Corrections meetings focused on prisoner treatment, COVID-19 protocols, and legislation. Meetings were led by a handful of California's criminal justice-oriented organizations, such as the Anti-Recidivism Coalition (ARC) and the Children's Defense Fund of California.

CALIFORNIA OCEAN PROTECTION COUNCIL
Intern, Summer 2019
Manager: Holly Wyer

- Drafted scientific briefings on coral reef vulnerability and the ecological impacts of toxic chemical pollutants such as oxybenzone. These briefings informed the legislative process leading to Assembly Bill 60, which now prohibits the sale of sunscreens with harmful chemical ingredients.

NEXTGEN AMERICA
Policy Intern, Summer 2019
Manager: Dave Weiskopf (Senior Policy Advisor)

- Prepared a report on the Central Valley Salinity and Nitrate Alternatives (CV-SALTS) coalition.

MENTORSHIP

E3BUDDIES MENTORING PROGRAM
E3B GRASS & Outreach Committee, August 2024 - present

- Graduate student group dedicated to providing mentorship for undergraduate students pursuing studies in Environmental Biology or Evolutionary Biology of the Human Species in the Ecology, Evolution, and Environmental Biology department at Columbia University. Coordinating graduate-undergraduate pairings, hosting monthly mentorship lunches, and organizing bimonthly professional development workshops.

WOMEN IN SCIENCE AT COLUMBIA
Columbia University, August 2024 - present

- Graduate student organization dedicated to the advancement of women and underrepresented groups in science, technology, engineering, and math (STEM) fields. Members are individuals who are pursuing or hold advanced degrees in the STEM fields. Coordinating graduate-undergraduate pairings across Columbia and Barnard, and additionally, volunteering for Girls' Science Day, a semesterly outreach event featuring hands-on experiments for middle school students.

NEW YORK ACADEMY OF SCIENCES (NYAS)
Global STEM Alliance: Afterschool STEM Mentoring Program, Spring 2023- Spring 2024
Coordinator: Danielle Mink-Bellizzi

- Mentored 5th and 6th graders at KIPP STAR – Elementary, on the NYAS Spring 2023 Curriculum: Hiding in Plain Light. The curriculum was divided into five topic areas: (1) components of light, (2) light and your brain, (3) light interactions, (4) circuits and solar energy, and (5) fractals.

HELLMAN SUMMER SCIENCE AND MATH PROGRAM
Hellman Fellow, Summer 2018

- Completed a rigorous two-and-a-half-week residential program in science and math, which included approximately 40 hours of class time interspersed with leadership development and team-building activities. During the Fall semester, Hellman fellows partnered with high schools within the greater Bay Area. Paired with Lighthouse Community Charter Public School (K-12), where I TA'd a high school biology course.

Addressed questions on homework and led students through labs designed to get them thinking about the body.

VOLUNTEER EXPERIENCES

SOAPBOX SCIENCE NYC
Central Park, New York, NY
Organizer: New York University

INVASIVE PLANT SPECIES REMOVAL
Riverside Park, New York, NY
Organizer(s): Upper West Side Run Club, Riverside Park Conservancy

SATURDAY SCIENCE
Mortimer B. Zuckerman Mind Brain Behavior Institute, New York, NY
Organizer: Jerome L. Green Science Center

BRAIN AWARENESS WEEK 2024
Harlem, New York, NY
Organizer: NYC Chapter of Society for Neuroscience (BraiNY)

ASTRONOMY DAY 2023
Montshire Museum of Science, Norwich, VT
Organizer: Dartmouth College

SKILLS

LABORATORY TECHNIQUES

- DNA/mRNA extraction, PCR, gel electrophoresis, micropipetting, weaning, ear notching/ear punching, bacterial culturing, magnetic bead DNA purification, (small) rodent handling, seed germination assays

INSTRUMENTATION

- qubit fluorometer, AES, centrifuge, Eppendorf repeater, multi vortexer, freeze-drier, microscope(s), autoclave

TECHNOLOGY

- GitHub, CLI/bash, Raspberry Pi, Microsoft Office, Excel, R, Python, GNU Image Manipulation (GIMP), Adobe Creative Cloud (i.e., Photoshop, Illustrator)

REFERENCES

Deren Eaton - de2356@columbia.edu

Andrés Bendesky - andres@bendeskylab.com

Len Pennachio - LAPennacchio@lbl.gov